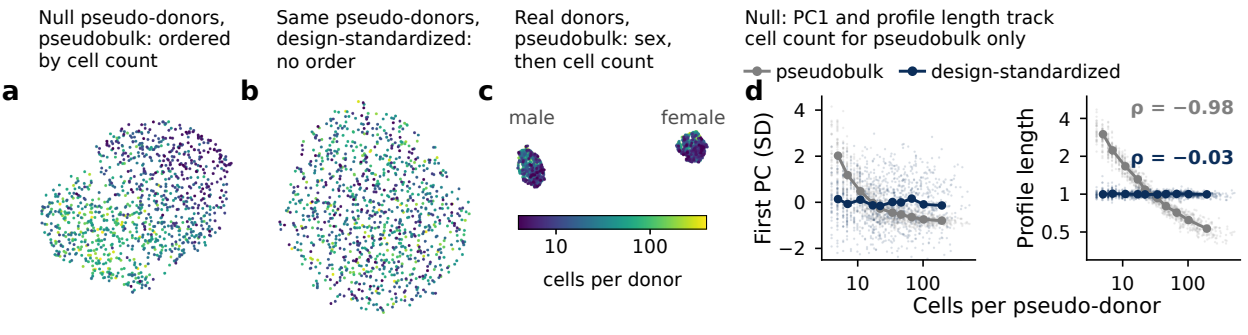

a
Null pseudo-donors, pseudobulk: ordered by cell count
b
Same pseudo-donors, design-standardized: no order
c
Real donors, pseudobulk: sex, then cell count
male
female
10
100
cells per donor
d
Null: PC1 and profile length track cell count for pseudobulk only
pseudobulk
design-standardized
First PC (SD)
4
2
0
−2
10
100
ρ = −0.98
ρ = −0.03
Profile length
4
2
1
0.5
10
100
Cells per pseudo-donor